

DAGs and the Backdoor Criterion: The Tutoring Example

Experimentation, A/B Testing and Causal Inference (SDA-DSC-213), SDAIA Academy. Handout for Day 3, Hour 1 (slides 87 to 92). All five Day 3 handouts use the same story: a tutoring programme at one school.

Step 1. The story

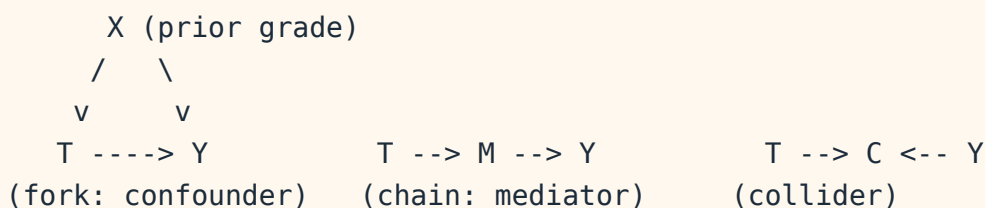
A school offers **after-school tutoring**. Students who attend score, on average, **4.5 points lower** on the final exam than students who do not. The head teacher asks: does tutoring hurt?

What we want **estimand: the ATE**: the average, over all students, of each student's exam score with tutoring minus the same student's score without it.

Step 2. Draw the assumptions

Before touching the numbers, write down what could cause what. Four variables:

Variable	Symbol	Role
Tutoring (yes or no)	T	treatment
Final exam score	Y	outcome
Prior grade (Low or High)	X	weak students are sent to tutoring, and weak students score lower anyway
Hours of practice	M	tutoring makes students practise more, and practice raises the score
Named on the honours list	C	decided from tutoring attendance and from the exam score



Step 3. The three structures (slide 89)

Structure	Shape	What conditioning on the middle node does	Rule
Fork	$T \leftarrow X \rightarrow Y$	closes the backdoor path; removes the bias	adjust for X

Chain	$T \rightarrow M \rightarrow Y$	blocks part of the causal effect itself	do not adjust for M
Collider	$T \rightarrow C \leftarrow Y$	opens a spurious association that was not there	do not adjust for C

Step 4. The backdoor criterion (slide 90)

A backdoor path is any path from T to Y that starts with an arrow **into** T. Here there is exactly one: $T \leftarrow X \rightarrow Y$. The rule: find a set Z that blocks every backdoor path and contains no descendant of T. $Z = \{X\}$ does it. M and C are both descendants of T, so they are out.

Backdoor paths from T to Y: $T \leftarrow X \rightarrow Y$ (one path) Adjustment set $Z = \{X\}$

Step 5. The numbers: naive versus adjusted

Eight students. X is the prior grade. Weak students are sent to tutoring more often.

Student	X (prior)	T (tutoring)	Y (score)
A	Low	1	60
B	Low	1	62
C	Low	1	64
D	Low	0	52
E	High	1	88
F	High	0	78
G	High	0	80
H	High	0	82

Naive comparison **estimator: difference in means**, ignoring X:

$$\text{mean}(Y \mid T=1) - \text{mean}(Y \mid T=0) = (60+62+64+88)/4 - (52+78+80+82)/4 = 68.5 - 73.0 = -4.5$$

Adjusted comparison: compare tutored and untutored students **within** each level of X, then average.

$$\begin{aligned} \text{Low: } & \text{mean}(Y \mid T=1, \text{Low}) - \text{mean}(Y \mid T=0, \text{Low}) = (60+62+64)/3 - 52 = 62 - 52 = +10 \\ \text{High: } & \text{mean}(Y \mid T=1, \text{High}) - \text{mean}(Y \mid T=0, \text{High}) = 88 - (78+80+82)/3 = 88 - 80 = +8 \end{aligned}$$

$$\text{ATE} = P(\text{Low}) \times 10 + P(\text{High}) \times 8 = (4/8) \times 10 + (4/8) \times 8 = +9$$

The estimate **estimate: +9**. Tutoring adds about 9 points; the naive -4.5 was the backdoor path $X \rightarrow T$ and $X \rightarrow Y$ talking, not tutoring.

Step 6. What goes wrong with the wrong controls

Adjust for M (hours of practice). Part of tutoring's effect works by making students practise. Holding practice fixed removes that part; the estimate falls toward the direct effect only, which answers a question nobody asked.

Adjust for C (honours list), or study only honours-list students. Among honours students, someone who was tutored needed a lower score to get on the list than someone who was not, so tutoring appears to lower scores. The association is manufactured by the selection. This is collider bias (slide 91): studying only hospital patients, only funded startups, only honours students.

Step 7. The procedure (slide 92)

1. Draw the DAG from domain knowledge before looking at correlations. 2. List the backdoor paths. 3. Choose Z that blocks them all, excluding mediators, colliders and anything measured after treatment. 4. Check every variable in Z is measured; an unmeasured confounder breaks the whole thing. 5. Record the graph with the analysis: it is the assumption the estimate rests on.

Three things to remember

1. The adjustment set comes from the graph, not from what improves the fit.
2. Confounder: adjust. Mediator or collider: do not.
3. Adjusting for X recovers the effect only if X is the whole story. That is the untestable bet, and Day 4 prices it with a sensitivity analysis.